PARTHA SINGHA ROY

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EDUCATION

Bachelor of Technology | *Electronics & Communication Engineering* — Aug. 2020– July 2024

Kalyani Govt. Engineering College — Kalyani, Nadia, West Bengal

WORK EXPERIENCE

Embedded System Engineer — Jul 2024 - Present

Aarish Technologies Inc. — New Town, Kolkata

- Developed a C++ (OOP-based) multi-threaded application using POSIX threads to run an audio CNN for Battlefield Surveillance Radar (BFSR) sound classification, and implemented both logarithmic and linear Automatic Gain Control (AGC) algorithms from scratch.
- Engineered a custom SDR communication system on PlutoPlus for real-time H.264 video and drone telemetry transmission, implementing OFDM-QPSK baseband processing using liquid-dsp and designing wideband frequency hopping (900 MHz–5.5 GHz).
- Developed an asynchronous smart vision system using C/C++ on Raspberry Pi and ESP32-C3, implementing YOLOv3-based object detection, RTP video streaming with embedded AI metadata, multi-sensor integration (GPS, PIR, MEMS mic), and optimized frame handling using a ping-pong buffer architecture.
- Developed a C++ framework for the Renesas RZ/V2L, enabling efficient execution of multiple CNN models on the DRP-AI accelerator through optimized and modular APIs.
- Engineered a custom Embedded Linux platform on Renesas RZ/V2L using Yocto, performing kernel customization, DTS development, and creation of a board-specific Yocto layer.
- Ported and integrated device drivers for IT6161 HDMI-MIPI bridge, IMX219/OV5647 camera sensors, and Wi-Fi modules.
- Spearheaded the end-to-end design and production of a custom RZ/V2L-based Single Board Computer (SBC), optimizing the Bill of Materials (BOM) for performance, reliability, and cost efficiency.

Software Engineering Intern — Jan 2023 – Jun 2023

FOSSEE, IIT BOMBAY — Remote

- Accomplished development of eSim software for Ubuntu 22.04(Porting KiCad v5 to KiCad v6).

Hardware Design Intern — Feb 2023 – April 2023

VLSI SYSTEM DESIGN, IIT BOMBAY — Remote

- Designed custom CMOS designs using SKY130 PDK, performing schematic capture (Xschem), layout (Magic/ALIGN), and post-layout characterization (NGSpice), including implementation of a 1-bit ADC and 130nm ring oscillator.

SKILLS

Languages: English, Bengali (Native), Hindi

Programming: C/C++, Python, Arduino, Bitbake, JavaScript, Bare Metal, OOP, CMake

Hardware: Custom Design RZV2L board, RZV2M Board, ESP32-c3, ESP32, Atmega 8/16/32, Pluto Plus

Tools: Altium Designer 24, KiCAD, EasyEDA, LTSpice, ESP-IDF, PlatformIO, Node-js, VS Code, Git, FFMPEG, GStreamer

Others: Uboot, Bootloader, Custom Linux kernel, Custom Linux OS

PROJECTS

DIGITAL SIGNAL PROCESSING IN ESP32 | C/C++, RTOS

Developed a dual-core ESP32-based real-time signal processing system implementing 16-bit FFT (Arduino FFT library) with core-level task segregation for deterministic DSP performance and seamless Wi-Fi data streaming to a web interface.

USBASP IN-CIRCUIT PROGRAMMER | Hardware Project

Developed a custom USBASP-based ISP programmer for Atmel AVR microcontrollers, optimizing firmware flashing workflows and achieving a 60% reduction in programming time.

COLLABORATIVE PROJECT BETWEEN AARISH TECHNOLOGIES, KGEC, AND JU

- Running CNN models (YOLOv2, YOLOv3, and HRNet) on Renesas RZ/V2M platform.
- Designed complete Single Board Computer (SBC) schematics for RZ/V2M, RZ/V2L, and RZ/V2H platforms using Altium Designer.